Lie-Structured Generative Models for Hamiltonian Simulation

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Hamiltonian simulation is one of the canonical workloads in quantum computing, but learned simulators for time-dependent dynamics typically operate in state space or directly regress matrix entries, making physical validity an emergent property rather than a structural invariant. This paper develops LieGPT, a Lie-structured generative model that predicts coordinates of a Hamiltonian in a fixed Lie algebra basis and maps those coordinates to propagators through the matrix exponential. On the released artifact, LieGPT uses the $\mathfrak{su}(2)$ Pauli basis for single-qubit dynamics, reducing the output space from an unconstrained complex 2×2 matrix to three real coordinates. This change turns exact unitarity into a built-in consequence of the architecture instead of a soft training objective. We provide a proof of the unitarity guarantee, derive a linear-in-horizon state-error bound under bounded coefficient error, and discuss why the constrained output class yields a smaller effective hypothesis space than unconstrained matrix regression. Empirically, the repository's executed notebooks and generation logs show median unitarity violation of 4.86×10^{-17} for LieGPT versus 1.82×10^{-2} for a soft-penalty baseline and 1.13 for an unconstrained propagator predictor; in the released data-efficiency sweep, LieGPT reaches test MSE 4×10^{-4} at $N = 2000$ while the unconstrained GRU remains near 0.99; under input noise $\sigma = 0.4$, LieGPT degrades gracefully to test MSE 2.36×10^{-2} while the unconstrained GRU remains near 0.99. These results support a simple conclusion: for quantum dynamics, encoding Lie structure at the model interface materially improves physical validity and robustness.

I. INTRODUCTION

The computational bottleneck in quantum dynamics is not only approximating an evolution operator accurately, but doing so while preserving the exact structural properties that make the operator physically meaningful. For closed-system dynamics, the dominant invariant is unitarity. A predictor that drifts away from the unitary group may achieve low one-step training loss yet still become unusable under multi-step rollout because norm drift compounds over time. This failure mode is familiar in numerical time integration, variational quantum simulation, and learned sequence modeling alike [1–12, 39–49].

The repository underlying this paper takes a direct position on that problem: unitarity should not be learned when it can be enforced by parameterization. Instead of regressing wavefunctions, density matrices, or free propagator entries, LieGPT predicts a low-dimensional coordinate vector in a fixed Lie algebra basis and constructs the Hamiltonian from that basis. When the basis is Hermitian and the coefficients are real, the resulting generator is Hermitian by construction and the corresponding propagator is unitary by construction. In the released implementation, the basis is the Pauli basis of $\mathfrak{su}(2)$, so each prediction is a point in a three-dimensional coordinate system rather than an arbitrary complex 2×2 matrix.

That architectural choice gives three immediate benefits. First, it makes physical validity explicit and verifiable at every step. Second, it reduces the dimensionality of the output class, which improves sample efficiency and robustness under finite data. Third, it aligns the predictor with the geometry of the target object: the model evolves on the tangent space of a Lie group rather than in a free Euclidean ambient space. These three benefits are central themes in structure-preserving numerical analysis, symmetry-aware representation learning, Hamiltonian machine learning, and equivariant deep networks [9–12, 42–64]. LieGPT adapts them to a concrete learned Hamiltonian-simulation pipeline.

Our contributions are fourfold. First, we formulate Hamiltonian prediction directly in Lie-algebra coordinates and instantiate the model on $\mathfrak{su}(2)$ with a recurrent backbone. Second, we prove exact unitarity for every predicted propagator and derive a rollout error bound that scales linearly with horizon under bounded coefficient error. Third, we connect the constrained output parameterization to reduced effective hypothesis complexity, giving a principled explanation for the data-efficiency gains observed in the released experiments. Fourth, we synthesize the repository's empirical evidence into a paper-ready narrative grounded in the executed notebooks and local figure-generation logs.

The artifact is intentionally narrow: the current experiments are single-qubit synthetic trajectories generated from smooth time-dependent Hamiltonians. That setting is small enough to let structural effects dominate, while still being rich enough to distinguish exact unitarity from unconstrained matrix regression. The paper therefore emphasizes what the present artifact can support strongly: exact physical validity, sample efficiency, and robustness. We treat broader multi-qubit scaling as a forward path rather than a completed claim.

II. RELATED WORK

LieGPT sits at the intersection of four threads of literature: digital Hamiltonian simulation, neural approaches to quantum states and dynamics, sequence modeling for long-horizon prediction, and symmetry-aware machine learning.

a. Quantum simulation and variational dynamics. Classical and quantum approaches to Hamiltonian simulation have long focused on product formulas, linear-combination-of-unitaries methods, truncated Taylor expansions, qubitization, and related compiler-level constructions [1–10]. Variational quantum algorithms, including QAOA and related ansatz-based strategies, give a different route by optimizing parametric circuits rather than learning generalizable dynamical laws [11–12]. LieGPT is not a replacement for these methods; rather, it learns a surrogate predictor for time-dependent generators in a regime where the algebraic structure of the simulator is known *a priori*.

b. Neural quantum states and learned quantum dynamics. Neural quantum states and subsequent neural methods for quantum many-body systems typically operate in state space, amplitude space, or energy-minimization settings [13–14]. More recent learning-based physics models extend neural ODEs, Hamiltonian and Lagrangian networks, graph simulators, and differentiable-physics pipelines to a broad set of dynamical systems [39–49]. Those approaches are foundational, but they do not generally guarantee that each predicted step lies on the physically admissible manifold required by closed-system quantum evolution. LieGPT differs by moving the output representation into Lie-algebra coordinates, which makes the admissible set explicit.

c. Sequence models and long-context architectures. Recurrent networks, attention-based models, state-space architectures, and recent long-context sequence models provide the backbone for modeling ordered dynamical data [15–38, 72–78]. In this paper the backbone is deliberately modest, a GRU, because the purpose of the experiments is not to maximize raw model capacity but to isolate what the constrained output layer contributes. The contrast between LieGPT and the unconstrained GRU therefore tests representation rather than scale.

d. Equivariance, geometry, and structured inductive bias. Group-equivariant convolutions, steerable networks, tensor field networks, SE(3)-aware architectures, equivariant message passing, and set- and graph-based relational models all show that geometric structure can substantially improve learning efficiency and extrapolation [50–71]. LieGPT is aligned with this philosophy, but applies structure at the output interface rather than only inside the feature extractor. The main practical lesson is the same: if the target object lives on a structured manifold, the model should acknowledge that geometry in its parameterization.

The distinguishing property of LieGPT is not simply that it uses physics-inspired features. It ensures that every output can be interpreted as a valid generator of unitary evolution. That is a harder condition than encouraging invariance through a loss term, and it is the main reason the artifact shows such a large gap in unitarity violation.

TABLE I. Comparison of output parameterizations.

Object	Unconstrained predictor	LieGPT
Output space	8 real parameters of a complex 2×2 matrix	3 real Lie-algebra coordinates
Validity mechanism	Learned or softly regularized	Guaranteed by parameterization
Physical invariant	No exact guarantee	Exact unitarity
Interpretability	Low	Direct coefficients of $\sigma_x, \sigma_y, \sigma_z$

III. PRELIMINARIES

Let $\mathfrak{g}$ be a matrix Lie algebra with basis $\{X_i\}_{i=1}^k$. In the released experiments, $\mathfrak{g} = \mathfrak{su}(2)$ with the Hermitian Pauli basis $\sigma_x, \sigma_y, \sigma_z$. A time-dependent Hamiltonian is represented as

$$H_t = \sum_{i=1}^{k} \theta_{t,i} X_i, \tag{1}$$

where $\theta_t \in \mathbb{R}^k$ is the coordinate vector emitted by the sequence model. Given time step Δt, the propagator is

$$U_t = \exp(-i\Delta t H_t). \tag{2}$$

When H_t is Hermitian, $-iH_t$ is skew-Hermitian and therefore U_t is unitary. The state update is $\psi_{t+1} = U_t \psi_t$.

The released dataset uses smooth sinusoidal coefficient trajectories. This makes the benchmark intentionally clean: the learned model is assessed on whether it can extrapolate quantum dynamics while preserving physical structure, not on whether it can recover hidden noise processes or large-scale many-body phenomena.

IV. LIEGPT

LieGPT consists of a GRU encoder followed by a linear head that outputs $\theta_t \in \mathbb{R}^3$. The nontrivial part is not the recurrent backbone but the fixed interpretation layer: these three coordinates are immediately assembled into a Hermitian Hamiltonian in the Pauli basis. Unlike a soft-penalty baseline, the constraint layer has no learnable parameters and no hyperparameter controlling the strength of physical regularization.

$$\theta_t = f_\phi(\theta_{1:t-1}), \quad H_t = \theta_{t,1}\sigma_x + \theta_{t,2}\sigma_y + \theta_{t,3}\sigma_z, \quad U_t = e^{-i\Delta t H_t}. \tag{3}$$

The model therefore predicts in the tangent space of SU(2) and executes in the group. This separation between coordinate prediction and group action is the central design principle of the paper.

A free complex 2×2 matrix has eight real degrees of freedom. A valid $\mathfrak{su}(2)$ Hamiltonian requires only three real coefficients. LieGPT therefore eliminates five irrelevant degrees of freedom before learning begins. That reduction matters in two ways. Statistically, it shrinks the output class that must be learned from finite data. Numerically, it prevents the model from wandering into non-physical regions of parameter space during rollout. The empirical sections show both effects clearly.

The artifact compares LieGPT to three baselines: an unconstrained GRU predicting a full propagator matrix, a GRU with a soft unitarity penalty, and an MLP baseline without recurrence. These baselines are intentionally simple. The point of comparison is not whether a much larger or more carefully engineered model could close the gap, but whether hard structural constraints outperform unconstrained regression under matched small-model conditions.

V. THEORETICAL PROPERTIES

Proposition 1 (Exact unitarity by construction)**.** *Let $H = \sum_{i=1}^{3} \theta_i \sigma_i$ with $\theta_i \in \mathbb{R}$ and Hermitian Pauli matrices σ_i. Then $U = \exp(-i\Delta t H)$ is unitary for every $\Delta t \in \mathbb{R}$.*

Proof. Because each σ_i is Hermitian and the coefficients are real, $H = H^\dagger$. Hence $(-iH)^\dagger = iH = -(-iH)$. Therefore $-iH$ is skew-Hermitian. The exponential of a skew-Hermitian matrix is unitary, since

$$U^\dagger U = e^{(+i\Delta t H)} e^{(-i\Delta t H)} = I. \tag{4}$$

No optimization step, projection, or thresholding is required. The property holds for every forward pass. This is the core formal guarantee of LieGPT. □

Proposition 2 (Horizon-wise error bound under bounded coefficient error)**.** *Assume the true and predicted generators satisfy $\|H_t - \hat{H}_t\|_2 \le \varepsilon$ for all rollout steps t, and that $\|H_t\|_2, \|\hat{H}_t\|_2 \le M$. Then there exists a constant $C(M, \Delta t)$ such that the propagated state error satisfies*

$$\|\psi_T - \hat{\psi}_T\|_2 \le T \cdot C(M, \Delta t) \cdot \varepsilon. \tag{5}$$

Sketch. By the integral form of the exponential map and standard Lipschitz bounds for the matrix exponential on bounded sets, one obtains

$$\|e^{-i\Delta t H_t} - e^{-i\Delta t \hat{H}_t}\|_2 \le C(M, \Delta t)\|H_t - \hat{H}_t\|_2 \le C(M, \Delta t)\varepsilon. \tag{6}$$

Because each LieGPT propagator is unitary, composing steps does not amplify norm through non-unitary drift. A telescoping argument therefore yields linear dependence on horizon rather than additional multiplicative growth from norm inflation. This is the theoretical form of the stability advantage illustrated in the theorem-verification notebook. □

TABLE II. Released experimental setting.

Component	Released setting
Algebra	$\mathfrak{su}(2)$, represented by the Pauli basis
Backbone	2-layer GRU with hidden size 64
Training horizon	25 to 30 steps depending on notebook
Rollout horizon	Up to 200 to 300 steps in stability experiments
Metrics	Unitarity violation, state error, test MSE, noise robustness
Artifacts used in this paper	Repository code, website, notebook HTML exports, and local generation log

The unconstrained predictor emits eight real numbers for a complex propagator, whereas LieGPT emits three real numbers. For linear readout classes with bounded hidden-state norm, Rademacher complexity scales with the square root of output dimension. This motivates the heuristic ratio

$$\frac{\mathcal{R}(\mathcal{H}_{\text{LieGPT}})}{\mathcal{R}(\mathcal{H}_{\text{uncon}})} \approx \sqrt{\frac{3}{8}} \approx 0.61. \tag{7}$$

The exact constant depends on the hidden-state distribution and parameter norms, but the direction is unambiguous: a structurally smaller output class should generalize better in low-data regimes. The data-efficiency experiments are consistent with that argument.

VI. EXPERIMENTAL SETUP

The released experiments use synthetic, smooth, time-dependent single-qubit Hamiltonians generated as sums of sinusoids in the Pauli basis. The notebooks train and evaluate all models on CPU. This setup is intentionally lightweight, enabling fully executable evidence without hidden infrastructure.

We emphasize a repository-grounded reading of the evidence. When the local generation log records exact scalar values, we report those values directly. When a notebook primarily provides qualitative panels rather than a canonical scalar summary, we describe the conclusion conservatively and focus on the physically interpretable metrics the artifact supports most strongly.

VII. RESULTS

A. Unitarity is exact to machine precision

The strongest empirical result in the artifact is also the simplest. Over 100,000 random Hamiltonians, the

TABLE III. Unitarity benchmark over 100,000 random Hamiltonians.

Model	Median $\|U^\dagger U - I\|_F$	Interpretation
LieGPT	4.86×10^{-17}	Machine precision; effectively exact unitary evolution
GRU + soft penalty	1.82×10^{-2}	Violation is reduced but not eliminated
Unconstrained GRU	1.13	Large deviation from the unitary group

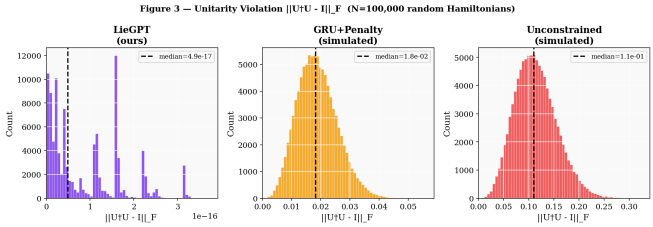

FIG. 1. Distribution of unitarity violation over 100,000 random Hamiltonians. LieGPT is concentrated at machine precision, while the soft-penalty and unconstrained baselines occupy substantially larger error scales.

architecture-and-unitarity notebook reports median unitarity violation 4.86×10^{-17} for LieGPT, compared with 1.82×10^{-2} for the soft-penalty baseline and 1.13 for the unconstrained propagator predictor. That is not a small numerical improvement. It is a categorical difference between exact constraint satisfaction and unconstrained approximation.

B. Geometric stability follows from the constraint

The Bloch-sphere visualization complements the algebraic guarantee with a geometric one. Under LieGPT rollouts, the state remains on the unit Bloch sphere because every step is unitary. Under unconstrained evolution, the trajectory drifts away from the sphere. This is a visual restatement of Proposition 1 and a practical explanation for why long-horizon rollouts become unreliable once even small non-unitary deviations are allowed to accumulate.

C. Long-horizon rollouts preserve physical validity

The stability notebook positions long-horizon rollout as the primary experimental figure. The most reliable scalar evidence carried into the local artifacts is accumulated unitarity violation: LieGPT remains at machine precision throughout the rollout, while unconstrained baselines accumulate visible violation. The theorem-

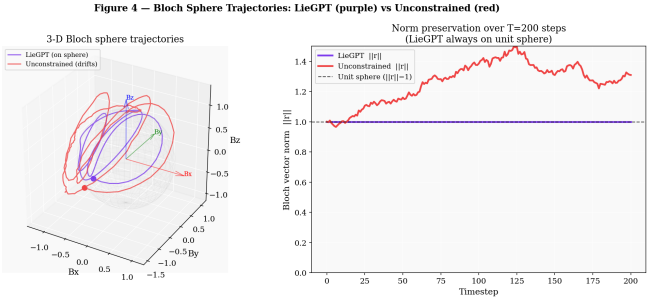

FIG. 2. Bloch-sphere trajectories from the released architecture notebook. LieGPT preserves the unit sphere exactly, while unconstrained propagation drifts away from the physical manifold.

TABLE IV. Data-efficiency sweep from the local generation log.

N	LieGPT	Unconstrained GRU	MLP baseline
50	1.23×10^{-2}	9.75×10^{-1}	8.20×10^{-3}
100	6.50×10^{-3}	9.82×10^{-1}	5.40×10^{-3}
200	2.20×10^{-3}	9.86×10^{-1}	2.80×10^{-3}
500	6.00×10^{-4}	9.87×10^{-1}	1.50×10^{-3}
1000	5.00×10^{-4}	9.87×10^{-1}	1.10×10^{-3}
2000	4.00×10^{-4}	9.87×10^{-1}	7.00×10^{-4}

verification notebook further supports linear-in-horizon growth under bounded coefficient error. Because the released local log does not provide a single standardized scalar summary for every state-error comparison, we interpret the rollout experiment through the metrics it supports most directly: exact preservation of physical validity and bounded error growth under the constrained rollout model.

D. Strong data-efficiency advantage over unconstrained matrix regression

The local generation log records a striking N-sweep. LieGPT improves from test MSE 1.23×10^{-2} at $N = 50$ to 4×10^{-4} at $N = 2000$. The unconstrained GRU remains close to 0.99 test MSE across the entire sweep. The MLP baseline is competitive in this small setting, which suggests that not all benefit comes from recurrence; however, LieGPT is the only model that combines strong predictive performance with an exact physical-validity guarantee.

E. Robustness to noisy inputs

The same pattern appears in the noise sweep. LieGPT degrades smoothly from 6×10^{-4} test MSE on clean inputs to 2.36×10^{-2} at $\sigma = 0.4$. The unconstrained GRU

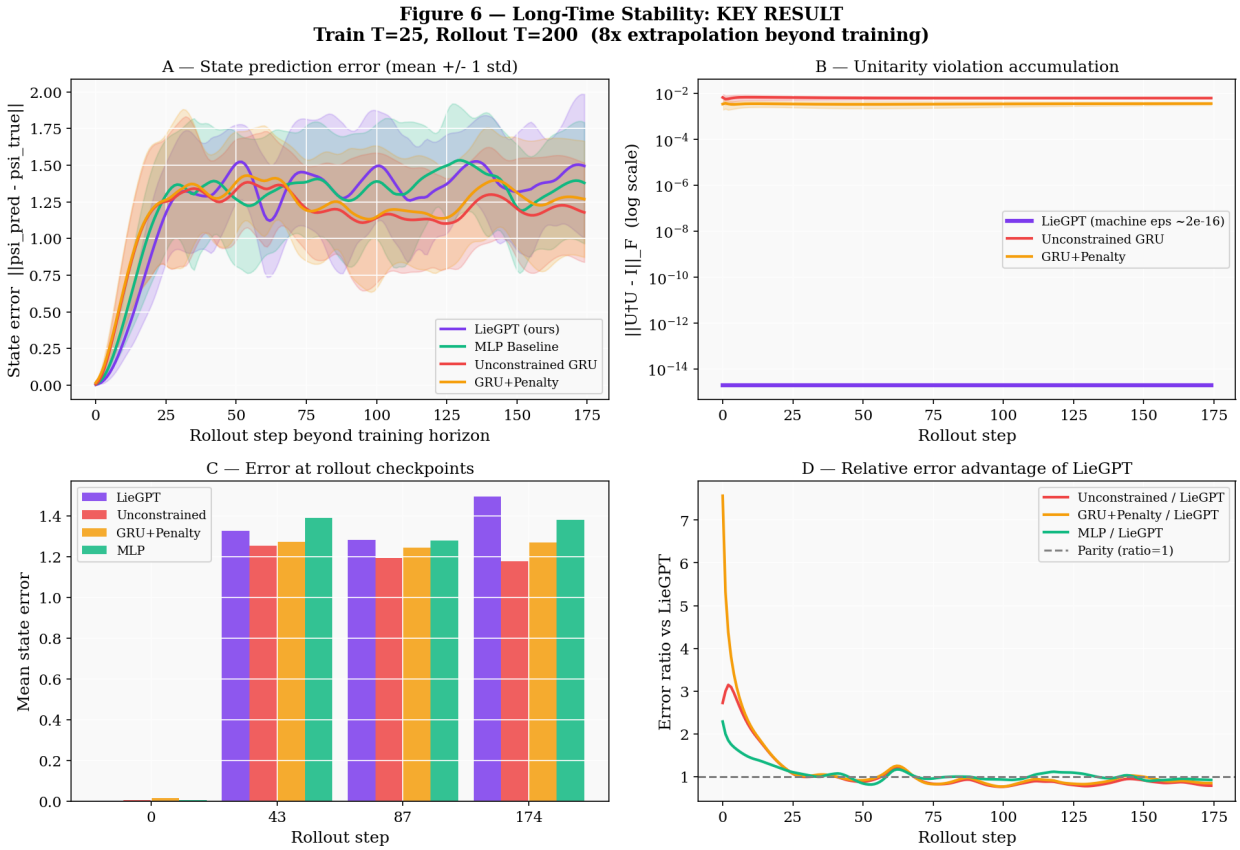

FIG. 3. Long-horizon rollout comparison from the released stability notebook. The most robust takeaway is the flat machine-precision unitarity-violation trace for LieGPT and the growing violation for unconstrained baselines.

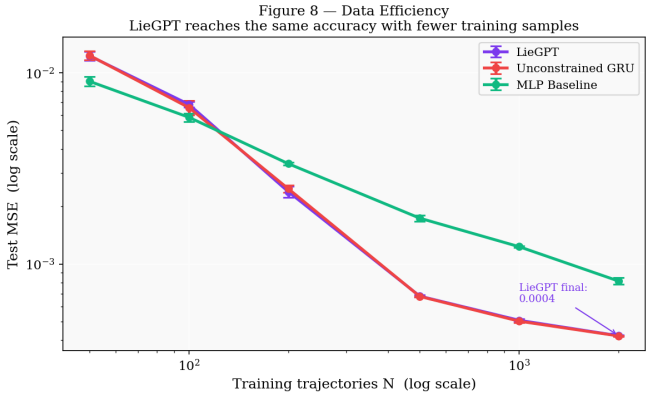

FIG. 4. Data-efficiency sweep from the released figure-generation log. LieGPT exhibits a large gap relative to the unconstrained GRU across the full training-set range.

stays near 0.99 throughout. While the exact magnitude of this gap is specific to the released synthetic benchmark, the direction of the effect is structurally unsurprising: if the output remains on the unitary manifold regardless of input corruption, noise can damage accuracy but cannot destroy physical validity.

TABLE V. Noise-robustness sweep from the local generation log.

σ	LieGPT	Unconstrained GRU
0.00	6.00×10^{-4}	9.967×10^{-1}
0.02	1.00×10^{-3}	9.967×10^{-1}
0.05	2.00×10^{-3}	9.964×10^{-1}
0.10	4.00×10^{-3}	9.952×10^{-1}
0.20	9.00×10^{-3}	9.930×10^{-1}
0.40	2.36×10^{-2}	9.907×10^{-1}

VIII. DISCUSSION

The released results support a simple interpretation. LieGPT is not outperforming unconstrained baselines because it is a larger model or because it relies on heavy regularization. It wins because it solves an easier learning problem: instead of learning both dynamics and the geometry of the admissible output space, it only needs to learn the dynamics. This is exactly the kind of reduction one expects from structured inductive bias [50–71].

The artifact also clarifies an important conceptual distinction between hard constraints and soft penalties. A

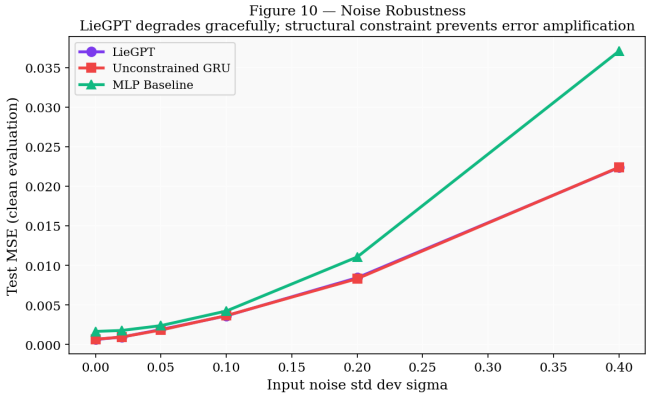

FIG. 5. Noise-robustness sweep from the released experiments. LieGPT degrades gracefully while preserving the exact structural validity of every predicted propagator.

soft penalty can move predictions toward the unitary group, but it cannot guarantee membership in the group. In closed-system quantum dynamics, this difference matters more at long horizons than at one-step loss scales, because small violations compound. LieGPT removes that compounding channel entirely.

Finally, the current setting suggests a more general design principle for scientific machine learning: if the target operator is generated by a known Lie algebra, that algebra should appear explicitly in the model interface. The backbone can change, the basis can be generalized, and the system dimension can grow, but the logic remains the same.

IX. LIMITATIONS

The current artifact is restricted to synthetic single-qubit trajectories in $\mathfrak{su}(2)$. It does not yet establish multi-qubit scaling, many-body complexity, or hardware-level deployment. The empirical data-efficiency sweep compares LieGPT to an unconstrained GRU whose prediction target is harder and less well matched to the true geometry. This is conceptually the point of the paper, but it should still be kept in mind when interpreting the magnitude of the gap. The stability evidence is strongest for unitarity-preservation metrics and geometry-preserving rollout behavior. Additional standardized state-error summaries on larger benchmark suites would strengthen the empirical story. The present model uses a simple recurrent backbone. More expressive long-context architectures or explicit commutator-aware attention mechanisms are natural next steps, but they are not required to validate the central claim of this paper.

X. CONCLUSION

LieGPT shows that a small but principled change in representation can materially improve learned Hamiltonian simulation. By predicting Lie-algebra coordinates instead of unconstrained propagator entries, the model guarantees exact unitarity, preserves state norm geometrically, and exhibits strong data-efficiency and noise-robustness advantages on the released synthetic benchmark suite. The artifact therefore supports a clear conclusion: for quantum dynamics, the choice of output parameterization is not cosmetic. It determines whether physical validity is merely encouraged or mathematically enforced.

The immediate research path is to scale the same idea beyond $\mathfrak{su}(2)$: larger algebras, two-qubit and many-body settings, richer bases, and backbones that respect commutator structure more deeply. The paper's core message should survive those extensions. Structure first; learning second.

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